

Does Income Affect Climbing the Energy Ladder? A New Utility-Based Approach for Measuring Energy Poverty

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ABSTRACT

Energy poverty measures are gradually becoming less relevant for fast-developing countries, where the energy mix consists of traditional and modern energies. We propose a new approach for measuring energy poverty by modifying the Exact Affine Stone Index (EASI) demand system to include implied disutility of energy use. The disutility arises from the effects of price or income changes and the use of polluting energies. Using data from Vietnam, we found that energy poverty could happen at higher income levels than the level considered in the literature, and higher incomes may not encourage households to climb the energy ladder. However, consuming carbon-intensive fuel does not necessarily mean energy poor.

Keywords: Energy poverty, Fuel poverty, Household energy use, EASI demand system, Vietnam, Household living standards survey

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1. INTRODUCTION

A broad consensus suggests that energy poverty is the lack of modern energy services because households do not have access to or cannot afford them. However, no single measure of energy poverty provides a reference point. Instead, past energy poverty measures mainly deal with one or several facets of energy-poor. They have provided a comparison across countries or regions and intertemporal differences that may help progress towards minimizing energy poverty. However, some concerns have arisen beyond the scopes of these studies. For example, the widespread use of a diversified residential energy mix consists of carbon-intensive fuels, even in wealthy households in the emerging economies. The underlying reason for having such an energy mix is living habits linked to the abundant availability of polluted energy at lower prices. Thus, practitioners and policymakers may fail to scrutinize the issue, subsidy programs may be wrongly targeted, and policies to relieve energy poverty may be ineffective.

In this paper, we develop a new method to evaluate energy poverty. This method is based on the disutility that occurs if households use polluted energy, e.g. solid biomass, or have to give up other demands to afford clean energy, e.g. electricity. For instance, a well-off family that can afford cleaner energy keeps consuming solid biomass as a long-term habit. The family absorbs much indoor pollution and derives disutility due to pollution. Another case is that a low-middle-income household pays a significant proportion of their disposable income for clean energy, so they must sacrifice other demands. We propose an extension of Lewbel and Pendakur's (2009) Exact Affine Stone Index

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(EASI) demand system to quantify implied disutility associated with consumer choices when using a different mix of energies, some with emissions. In an application, we adopt three waves of the Vietnam Household Living Standards Survey (VHLS) data (2012, 2014 and 2016) with more than 24,000 observations.

The empirical results demonstrate that the new method is more comprehensive than the previous approaches to identifying energy poverty households in many unsolved circumstances. Our findings suggest that energy poverty can be escalated at different income levels, including middle-income or above levels. Furthermore, we also find that consuming conventional fuels, usually considered dirty energy, does not always mean the household is in energy poverty. Thus, energy policy also needs to be updated to adapt to these new conditions.

In the literature, common energy poverty measures are mainly based on two pillars, *accessibility* to modern energy like electricity and *affordability* measured in terms of energy expenditure. Accessibility measures implicitly assume that households gradually move away from traditional energy inputs like solid biomass as countries progress in the development path, a concept proposed by the International Energy Agency (IEA, 2002). Accordingly, a household is in energy poverty if it lacks access to modern energy services or depends on traditional biomass for activities, such as cooking meals.

Some examples of studies that have used IEA measures are Sesan (2012), Bhide and Monroy (2011), Mirza and Szirmai (2010), Birol (2007), and Sagar (2005). These researchers identified the reliance on traditional biomass in cooking as a measure of energy poverty. Meanwhile, studies by Singh and Inglesi-Lotz (2021), Legros et al. (2009) and Foster et al. (2000) provided examples that consider the accessibility to electricity as a proxy for energy poverty. Other studies (Pereira et al., 2011; Action, 2010; Pachauri et al., 2004; Foster et al., 2000; Goldemberg et al., 1988) defined energy poverty by estimating the minimum level of energy needed to maintain a certain living standard.

Within the accessibility perspective, energy poverty could also be evaluated by composite measures, which are combinations of several weighted indicators. Two well-known composite measures are the Energy Development Index (EDI) developed by IEA (2010) and the affordable and clean energy goal (Sustainable Development Goal 7 [SDG7]) developed by the United Nations Desa (2016). Both the EDI and the SDG7 rely on electricity accessibility and macro indicators, such as the average usage level of modern fuels in the residential sector or renewable share in total final energy consumption. Thus, these methods are more suitable for a cross-country comparison than a household-level analysis.

Providing a composite measure for a specific country, Nussbaumer et al. (2012) proposed the Multidimensional Energy Poverty Index (MEPI), which focuses on a household's privation to modern energy services. This method considers a set of micro-level indicators that includes: (a) the use of non-modern fuels, (b) cooking produces indoor pollution, (c) having access to electricity, and (d) having either a fridge or a radio/television or a landline/mobile phone. A recent study by Delugas and Brau (2021) applied the MEPI method to identify a significant link between subjective well-being and energy poverty intensity. In applications of the MEPI, Munyanyi and Churchill (2022) found that foreign aid helped reduce energy poverty by 3.3% in Senegal. In addition, Abbas et al. (2020) suggested that better socioeconomic status mitigates the energy poverty in South Asia. Meanwhile, studies by Dong et al. (2022), Dong et al. (2021) and Zhao et al. (2021) all employed an energy poverty composite index with four components: energy service availability, energy consumption cleanliness, energy management completeness, and energy affordability and efficiency.

The second pillar, affordability, was initially introduced to measure fuel poverty by Lewis (1982) and considered the inability to afford warmth for the house. Later on, Boardman (1991) defined fuel poverty as if a household has energy costs higher than 10% of their income to maintain

adequate indoor temperature. A study by Phimister et al. (2015) adapted the 10% indicator with the consideration of disposable income, which excludes housing costs instead of total household income. Other studies by Barnes et al. (2011) and Pachauri et al. (2004) adopted a 10% indicator to analyze energy poverty, instead of fuel poverty, for India and Bangladesh. The 10% measurement was officially applied by the U.K. Government in their formal legislation before Hills (2011) proposed a new definition of fuel poverty as low-income high cost (LIHC). According to the LIHC, a household is in fuel poverty if its income is lower than 60% of the national median income level while its energy expenditure is above the median level. More recently, studies by Dogan et al. (2022), Nguyen et al. (2019) and Mayer et al. (2014) adopted LIHC to determine households in energy poverty. Besides, among several studies that applied both the 10% indicator and LIHC, Charlier and Kahouli (2019) and Welsch and Biermann (2017) found that more energy poverty leads to higher energy price elasticity.

The terms energy poverty and fuel poverty have been used interchangeably in the literature. The studies of Charlier and Kahouli (2019), Romero et al. (2018), Welsch and Biermann (2017), Herrero (2017), Schuessler (2014), and Foster et al. (2000) are some examples among many. It may imply that the boundary between energy poverty and fuel poverty is getting blurred, such as the issue of households in many developing countries is no more about inaccessibility but how to afford the energy bill. Furthermore, while affordability has become a more common concern than accessibility, the expenditure-based approaches could not become dominant in measuring energy poverty because they ignore the mitigating expenditure behavior (Phimister et al., 2015). Some recent studies, including Pérez et al. (2021), Charlier and Legendre (2019), Welsch and Biermann (2017), and (Phimister et al., 2015), paid more attention to household behavior by involving subjective measures together with the affordability. However, subjective measures could also not be consistent solutions because they rely on the household's perspective, in which families living in poor condition houses are more likely to address themselves as energy poor.

To this end, extant energy poverty measures are gradually becoming irrelevant for rapidly developing countries, such as China or Vietnam. On the one hand, the accessibility approach has limited relevance in those countries as the accessibility to modern energies is no longer an obstacle and solid biomass is still a significant part of the energy used for cooking. This situation violates the implicit assumption of the accessibility measure, that people substitute low quality, high pollutant energy sources with modern energy as they escape income poverty. Instead, it could be associated with the energy consumption behaviors and living habits linked to the abundant availability of such fuels at lower prices. Thus, neither traditional methods entirely capture the energy poverty in these countries nor comprehensively accommodate the choice to use solid biomass associated with the customer behavior perspective.

On the other hand, the affordability methods have also been questioned in the context of rapidly developing countries for various limitations. For example, Li et al. (2014) recommended that these measures could only be used under specific conditions. Those conditions include households located in cold climate regions, households lacking access to modern energy (for cooking), and households having difficulty achieving satisfactory indoor temperatures with reasonable energy costs. Furthermore, the 10% indicator is debated in the literature by Schuessler (2014) and Heindl (2014) as inappropriate for countries other than the United Kingdom because of cross country parity and cost of living differences. The LIHC definition is also disputed by Moore (2012) and Boardman (2012) because of several deficiencies, such as the potential exclusion of poor households living in small houses and the hidden impact of escalating fuel prices. Further discussions on the advantages and disadvantages of those methods can be found in Romero et al. (2018).

In this paper, we proposed a new quantitative method that could potentially resolve some caveats of the current energy poverty measures. The approach could be well matched with situations where countries use a combination of conventional and modern energies. Vietnam, one of the most rapidly developing countries of the 21st century, can be examined as a typical example. The economic development has led the poverty headcount ratio (at US\$1.9 a day) to reduce drastically from 38% in 2002 to 1.9% in 2018. The electrification rate of Vietnam reached 100% in 2017, and the rate of households with durable equipment was 99.7% in 2016. The proportion of households that currently use liquified petroleum gas (LPG) in cooking in 2016 was more than 75% (WorldBank, 2019). Furthermore, the rate of solid biomass and waste used in the residential sector's total energy consumption reduced dramatically from more than 84% in 2000 to approximately 33% in 2017 (IEA, 2020).

Based on the above data, current energy poverty measures are unable to assess household energy poverty in Vietnam. First, the accessibility approach could either overlook or overcount the percentage of households in energy poverty, thus producing conflicting results. On the one hand, the accessibility to modern energy could potentially underestimate Vietnam's actual energy poverty because the entire population has access to electricity, and a greater proportion uses LPG. On the other hand, given that a significant proportion of the population still uses traditional fuel, it is likely to overestimate energy poverty. Evidence exists that many Vietnamese households use solid biomass as a traditional practice for cooking because of the abundance of supply, although they have access to modern energies. For example, even in big cities, while LPG is commonly used for cooking, people occasionally continue to use solid biomass. These households may not necessarily come from the low-income group nor have difficulty affording modern energy but habitually continue to use solid biomass sporadically. It is difficult to distinguish them from other households with mitigating expenditure behavior that uses traditional fuel to minimize the expenses.

Even a multidimensional energy poverty index, such as MEPI, does not provide better estimates of Vietnam's energy poverty. The underlying reason is that most of the attributes used in calculating these measures are equal and unvarying for all households. For example, all households have access to electricity, and the majority use LPG (75%) and possess durable equipment (99.7%). For this reason, the dimensions of MEPI finally collapse to a single dimension, i.e., whether or not the household uses solid biomass or fuel that causes pollution. As a result, this measure may report exceptionally low estimates for energy poverty.

Second, affordability approaches could also produce conflicting estimates for energy poverty. Initially, these methods were developed to suit households in developed countries that use only modern energies. Thus, these measures are not fully applicable to countries with a mixture of modern and traditional energy. For instance, many households in richer cohorts use traditional, low-quality energies, but they are strictly excluded as a pre-setting of the model. We argue that a well-off family may also need to be considered energy-poor if their health is at risk due to consuming too much low-quality fuel. In low-income households, the low price of traditional energy may make the energy budget share too small to be counted in energy poverty, even if a family is in financial hardship and highly depends on pollutant fuels. However, if a household has income closer to the pre-determined threshold and uses modern energy, the household will have a relatively higher expenditure share to be counted as energy-poor.

To the best of our knowledge, no extant literature has investigated energy poverty considering households' energy choice behavior in response to varying prices and pollution preferences to maximize their utility. A study by Pérez et al. (2021) interested in consumption behavior in cooking fuels, but this study did not address the impact of energy prices and emissions. Our study aims to fill

this gap by introducing an extension to the EASI demand system proposed by Lewbel and Pendakur (2009). We refer to our version of this model as the EASI-E demand system and argue it is appropriate for different economic situations as long as the energy mix consists of traditional and modern energies.

The rest of the paper is structured as follows. Section 2 explains the idea of implied disutility from an energy consuming behavior's point of view and suggests a new definition for energy poverty. Section 3 proposes the extension of Lewbel and Pendakur's (2009) EASI demand system to quantify the implied disutility and identify the energy-poor households. Section 4 describes Vietnamese data and background information in an application of the new method. Section 5 provides estimates and findings. Section 6 concludes the paper and discusses policy implications.

2. IMPLIED DISUTILITY OF ENERGY USE AND A NEW DEFINITION OF ENERGY POVERTY

Energy consumption behavior or consumer choice is a crucial factor but is often overlooked in recent energy poverty measurements. Without understanding consumers' choices, accessibility methods may fail to explain why households keep using traditional biomass when they have excellent access to electricity and other modern energies. Similarly, affordability methods may be unable to determine if a home has naturally low energy expenditure, i.e. not energy poverty, or because it reduces energy spending compulsorily, i.e. energy poverty.

In this section, we clarify the unique characteristics of energy-consumption behavior. Then, we discuss potential consumer choices with the corresponding implied disutilities that evoke a new definition of energy poverty. These discussions purposefully target developing countries with a diverse energy mix, including traditional biomass and modern energies. However, they are also viable for households in developed countries where the energy mix consists of varied sources.

2.1 Unique characteristics of energy-consuming behavior

To evaluate energy use behavior, we need to define critical characteristics that distinguish it from other goods' consumption behavior. Specifically, they include (i) fractional and fixed useful output, (ii) involving emission-related trade-offs, and (iii) implied utility as a response to indoor emission.

i) Fractional and fixed useful output energy

Only a fraction of the used energy is transferred or transformed according to the intended use, so-called *useful output energy*, such as the portion of the electricity used for lighting. The energy that does not contribute to lighting, for example, heating the light bulb, is called wasted energy and not accounted for useful output energy. Often, that fractional useful output energy is fixed in the sense that it does not vary in the short term. For more details, a particular family will have a specific need for energy for daily activities regardless of the energy type, e.g. the useful output energy requirement for cooking meals is the same whether using gas or electricity. If the relative prices of different energies change, households tend to switch to cheaper alternatives but stick with the current useful output energy. Even if energy prices change entirely, families are unlikely to be able to adjust their useful output energy promptly. For instance, the useful energy needed to boil the same amount of potatoes for food demand does not change whether energy prices rise or fall. The fixed useful output energy property is similar to the well-known price inelasticity property, but with more consideration given to the consumer's flexibility to change energy types.

ii) Emission-related trade-offs

In choosing a bundle of non-energy goods, consumers often need to consider the only trade-off of price and utility. However, with energy products, savvy consumers need to carefully consider the exchange between utility, price, and indoor emissions, which will impact upon family health. Emissions related trade-offs may not be apparent in the case of developed countries because of the strict regulations on the energy types allowed for household use. But, it is easily seen in developing countries, where the energy mix is more diversified with traditional biomass, gas, and electricity. In addition, households have excellent access to all kinds of energy, and many of them can switch to using alternative energy with almost no barrier. For example, many families in Vietnam own electric stoves, gas stoves, and wood-fired stoves to utilize for cooking meals whenever they want. Among energies, traditional biomass is relatively cheaper than other energy sources but is associated with higher indoor pollution levels.^{1,2} As a result, households who use traditional biomass will face a lower energy expenditure but may encounter more health problems as a trade-off.

iii) Implied utility as a response to indoor emission

Health problems associated with indoor-polluted energy can be encountered quickly but can also take years to become apparent, depending on health status and intensity of use. Thus, it may not be recognizable to many families or may be overlooked if traditional cooking fuels are only occasionally used. From the consumer behavior perspective, those households may enjoy higher utility, i.e. be happier, with a smaller energy cost. However, they do not realize that their utility should be lower because they have to pay back in terms of their health costs later. On the other hand, a knowledgeable household may want to replace traditional biomass with electricity and other modern energies to avoid health risks. Although the utility reduces because the energy cost becomes more expensive, this choice still makes the households feel happy, i.e. higher utility than measured. Therefore, we define the unperceived level of households' utility, either by consuming clean or dirty energy, as an *implied utility*. More broadly, the implied utility can be effectively applied to explain many other unsolvable or only partly answered cases in the literature on energy poverty. For example, by finding implied utility, we can distinguish the households who need to reduce their energy bill compulsorily from the homes with the same income level but naturally need small energy demands.

2.2 From implied utility to a new definition of energy poverty

Drawing from the above discussion, we graphically illustrate the consumer behavior of energy use in this section to propose a new measure of energy poverty. This measure includes both traditional definitions and solutions for unexplained gaps in the previous methods.

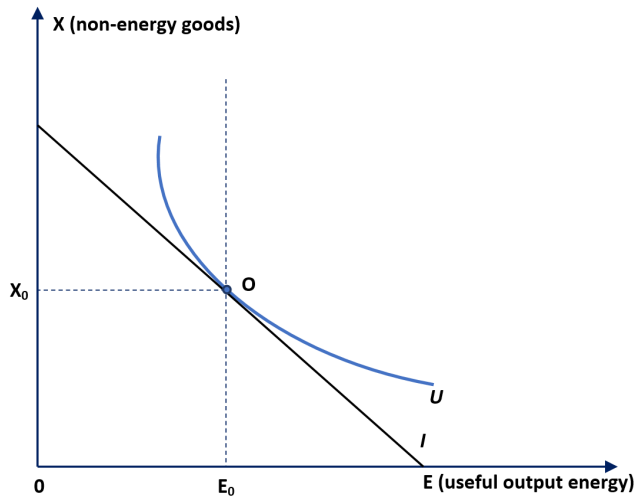
Consider the following economic model. Assuming a typical household consumes two bundles of goods, of which X is a bundle of ordinary goods, e.g. food and clothing, and E is a bundle of energies, including electricity, gas, and traditional fuels. According to conventional consumer behavior theory, given prices of X and E , a household's budget constraint I and its preferences U (or utility function) can determine optimal bundles of X and E , which is the point $O (E_0, X_0)$ in Figure 1.

1. The term "cheaper" here means not only the unit price but also the total energy cost of using traditional biomass is smaller than total cost of using electricity or other modern energies, regardless of the big difference in energy conversion efficiency. The reason of this could be an overabundance in the traditional biomass supply that even households can amass on their own.

2. This pollution referred to here is indoor pollution only as the model consider the consumption of energies by the households for their daily activities.

Point O is theoretically determined by budget constraints, prices, and each bundle's expected consumption. Value E_0 in Figure 1 demonstrates the useful output energy discussed earlier.

Figure 1: Theoretical Consumer Choice of Normal Goods (X) and Useful Output Energy (E)



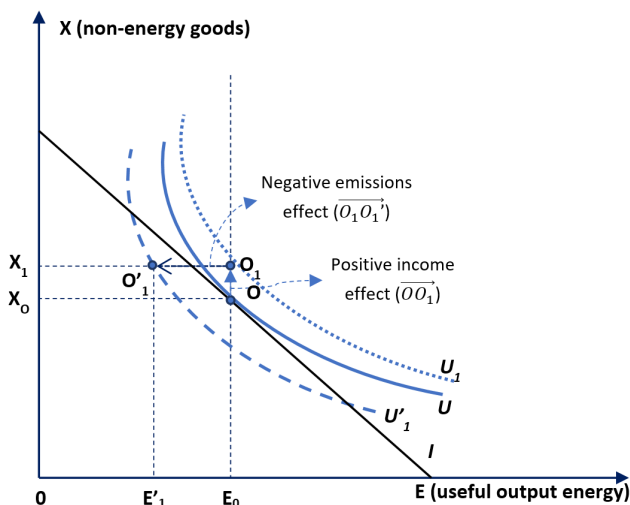
Notes: This figure depicts the authors' idea of adapting a theoretical model of consumer behavior. The optimal choice is point O given the utility curve, budget, and two goods bundles: non-energy goods and useful output energy. Because the useful output energy is sticky in the short term, any change in the consumer's choice will move vertically along the line through E_0 .

The useful output energy E_0 in this model can be obtained by having either one type of energy, e.g. electricity only, or a mix of energies, e.g. biomass and electricity. For simplicity, we classify the energy bundle E into two types based on the pollution emitted. The modern energies consisting of electricity and gas have no or low emissions, and conventional energies consisting of all traditional energy, such as charcoal, solid biomass, and kerosene, with considerable emissions. Here, we explain a household's choice to change the energy bundle but retain the same useful energy output (E_0) by changing the bundle's composition. There are two possibilities:

Case I: The use of conventional energies in the energy mix increases (with decreasing modern energies) to remain in the same useful energy output (E_0).

Case II: The use of modern energies in the energy mix increases (with decreasing conventional energies) to remain in the same useful energy output (E_0).

Figure 2 illustrates the first case. The household that uses more conventional energies has a bigger remaining budget to consume more bundle X because conventional energies are usually cheaper. Thus, the utility initially increases from level U to a new level U_1 with a new amount of normal goods bundle X_1 ($X_1 > X_0$), while the useful energy output (E_0) remains unchanged. The move from O to O_1 resembles a positive income effect of the price change as it moves from one utility level to another higher utility level. However, the more conventional energies consumed, the higher the pollution level generated. Thus, this choice will eventually have negative impacts, e.g. household health problems, which we call the negative emission effect. Because that emission effect exists despite not always apparent, we can identify that the household's utility implicitly drops to a lower level of U'_1 , at point O'_1 . The utility at O'_1 is similar to a household that consumes the same amount of X but smaller bundle E ($E'_1 < E_0$). The new utility level U'_1 , which is an implied utility, could be higher or lower than the initial utility U , depending on the magnitude of the emission and income effects.

Figure 2: The Consumer Choice to Consume More Conventional Energies

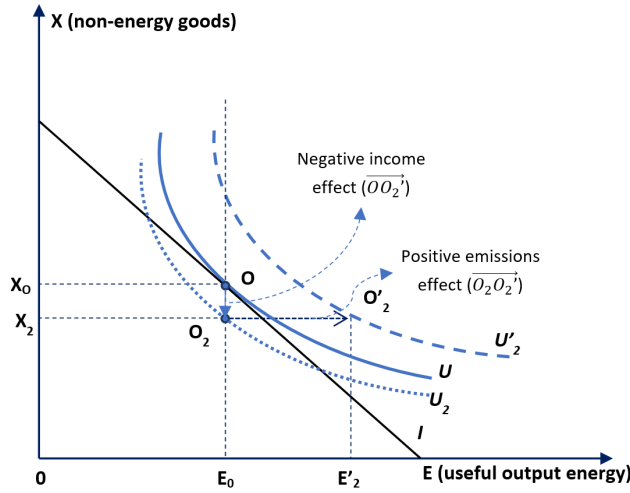
Notes: This figure depicts the authors' idea of implied utility when a household chooses to consume more conventional energies, e.g. solid biomass. Because conventional energies are less expensive, regardless of the efficiency, the family can reach a new optimal point, O_1 , with higher utility, $U_1 > U$, under the positive income effect. However, conventional energies are associated with more indoor pollutants, so the household experience a negative emissions effect. Their utility, reflected by implied utility U'_1 , is lower than U_1 but not necessarily lower than U . The household is considered energy poverty if $U_1 < U$.

Figure 3 demonstrates the second case, in which a household chooses to use more modern energy and pay a higher energy cost. Thus, the household's choice will move vertically to a point O_2 , which represents a bundle with less amount of X . This movement is a negative income effect caused by more expensive modern energy. At point O_2 , the utility was reduced because they only consume less of bundle X ($X_2 < X_0$, so $U_2 < U$). In this case, however, consumers can gain some indirect form of utility due to the absence of pollution and move to a higher utility. In other words, the analogous positive emission effect will implicitly increase the utility level from U_2 to a new indirect level U'_2 ($U'_2 > U_2$). Once again, U'_2 could be higher or lower than U , depending on the magnitudes of the emission and income effects.

In summary, households always have an implied utility by either consuming more conventional energies or navigating to use more modern energies. Depending on the magnitude of the income and emission effects, households could have different utilities, either higher or lower than the initial utility level. In this paper, we are interested in the possibility that families have lower implied utility, i.e., have disutility, which puts them at risk of energy poverty. The two cases discussed above show that the disutility can be identified if the net value of emission and income effects is less than zero. More precisely, the disutility happens when either:

- (i) the negative emission effect > the positive income effect in case I, or
- (ii) the positive emission effect < the negative income effect in case II.

This enables a simplified energy poverty measure that a household becomes more energy-poor if its energy choice causes disutility. We can also determine a standard reference for a country that the energy poverty line refers to as the national average utility. More specifically, we propose a simple but new definition of energy poverty: a household will be energy-poor if its implied utility is lower than the national average utility. This definition can cover accessibility methods because it can determine the energy poverty from using polluted energies. In addition, it covers affordability measures by identifying households paying too much for energy while or sacrificing other needs

Figure 3: The Consume Choice to Consume More Modern Energies

Notes: This figure depicts the authors' idea of implied utility when a household chooses to consume more modern energies, e.g. electricity. Because more modern energies mean more expensive, regardless of the efficiency, the family has a new optimal point, O_2 , with lower utility, $U_2 < U$, under the negative income effect. However, households who choose to use clean energies usually value their health and the surrounding environment highly. Thus, their utility, reflected by implied utility U_2' , is higher than U_2 but not necessarily higher than U . The household is considered energy poverty if $U_2' < U$.

for energy as energy poverty. Furthermore, this definition could overcome the drawback of strong assumptions in some other poverty measures. Specifically, households with conventional energies or considerably large energy expenditure share are not in energy poverty, as long as their implied utility level is not lower than the energy poverty level.

3. EASI-E: A DEMAND SYSTEM TO EVALUATE ENERGY POVERTY

In this section, we estimate energy poverty using our new definition based on the implied (dis)utility of energy use. Our estimation is based on Lewbel and Pendakur's (2009) Exact Affine Stone Index (EASI) demand system. The EASI system is powerful in estimating budget share and utility, flexible in setting and able to overcome many disadvantages from other demand systems (more details in Lewbel and Pendakur [2009] and Pendakur [2009]). We propose an extension to the EASI demand system that captures the consumers' energy use behavior when different energies have different prices and pollution levels. Thus, this extension enables the estimation of implied utility. We name this model the Exact Affine Stone Index features energy use demand system (EASI-E).

Following the EASI system procedure, we start with the cost function as in the original EASI model. The cost function yields:

$$\ln C(\mathbf{p}, u, \mathbf{z}, \varepsilon) = u + \sum_{j=1}^J m_j(u, \mathbf{z}) \ln p_j + \frac{1}{2} \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k + \sum_{j=1}^J \varepsilon_j \ln p_j. \quad (1)$$

In the above equation, $C(\mathbf{p}, u, \mathbf{z}, \varepsilon)$ is the minimum total expenditure subject to the target utility level u when the consumer is facing the vector of prices $\mathbf{p}$; $\mathbf{z}$ is a vector of observable characteristics; ε is a vector of unobserved characteristics; subscripts j and k are identifications of two different goods in a set of all household goods $\mathbf{J}$; $m_j(\cdot)$ and $a_{jk}(\cdot)$ are functions that will be presumed later in Eq. 7 and Eq. 8. The actual total expenditure, $x = C(\mathbf{p}, u, \mathbf{z}, \varepsilon)$, and budget shares, $w_j = \omega_j(\mathbf{p}, u, \mathbf{z}, \varepsilon)$, are observed from the data.

We modify this cost function to include an additional term representing the implied opportunity costs of choosing a different energy mix while remaining in the same useful output energy (E_0). From the explanation in the previous section, the opportunity cost could be a loss if the household experiences disutility or a gain if the family is better off in terms of implied utility. In line with this, we define this opportunity cost as the implied emissions cost as it is associated with the energy emissions. How large the implied emission cost depends on the following variables, (a) the emission level of the fuel and its alternative, which is expected to have exponential correlations (Dietz and Venmans, 2019, Golosov et al., 2014); (b) the prices of the fuel and its alternative, which is expected to have linear correlations; and (c) current level of utility, which is expected to have linear correlations. The total implied emission cost can be written as,

$$EC(\mathbf{p}, u, \mathbf{e}, \gamma) = \frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_j \ln p_k, \quad (2)$$

where subscripts j and k are identifications of fuel j and its alternative k in the energy mix E (E is a subset of all household goods J , $E \subset J$); and $\mathbf{e}$ and γ are vectors of consumption and emission rates of consumed energies, respectively. The term $\phi_{jk}(\mathbf{e}, \gamma)$ reflects the substitution effect caused by emissions, in which the household considers the demand of energies in the energy mix E based on their emissions. For simplicity, we suggest the form of $\phi_{jk}(\mathbf{e}, \gamma)$ as follows,

$$\phi_{jk}(\mathbf{e}, \gamma) = \varphi_{jk} e_j \exp(\gamma_j) e_k \exp(\gamma_k), \quad (3)$$

where $e_1, e_2, \dots, e_E \in [0; 1]$ are shares of fuels in total energy consumed, measured by the ratio of each fuel's expense to the household's total energy expenditure; $\gamma_1, \gamma_2, \dots, \gamma_E$ are emission rates of each energy; parameters $\varphi_{jk} = \varphi_{kj}$ for all $j \neq k$ and $j, k \in 1, 2, \dots, E$.

We can accommodate the total implied emission cost $EC(\mathbf{p}, u, \mathbf{e}, \gamma)$ from Eq. 2 into Eq. 1 to obtain a new cost function as follows:

$$\begin{aligned} \ln C(\mathbf{p}, u, \mathbf{z}, \mathbf{e}, \gamma, \varepsilon) = & u + \sum_{j=1}^J m_j(u, \mathbf{z}) \ln p_j + \frac{1}{2} \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k \\ & + \frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_j \ln p_k + \sum_{j=1}^J \varepsilon_j \ln p_j. \end{aligned} \quad (4)$$

Following Shephard's Lemma, Hicksian budget share functions are equal to the price elasticity of the cost function. Thus, taking the derivative of the new cost function with respect to each price, we can obtain the new Hicksian budget-share function,

$$\omega_j(\mathbf{p}, u, \mathbf{z}, \mathbf{e}, \gamma, \varepsilon) = m_j(u, \mathbf{z}) + \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_k + \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_k + \varepsilon_j. \quad (5)$$

where $a_{jk}(\mathbf{z}) = a_{kj}(\mathbf{z})$ for all $j \neq k$. The difference in Eq. 5 compared to the budget-share function in Lewbel and Pendakur (2009) is the appearance of $\sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_k$ which can be considered as implied impact of emissions on energy-budget share, so-called the implied emission expenditure share.

With budget shares $w_j = \omega_j(\mathbf{p}, u, \mathbf{z}, \varepsilon)$ observed from the data, Eq. 5 can be rewritten to become Eq. 6 as follows,

$$w_j = \omega_j(\mathbf{p}, u, \mathbf{z}, \mathbf{e}, \gamma, \varepsilon) = m_j(u, \mathbf{z}) + \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_k + \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_k + \varepsilon_j. \quad (6)$$

Following the same process provided in the study by Pendakur (2009), we parameterize $m_j(u, \mathbf{z})$ and $a_{jk}(\mathbf{z})$ with additive structures, of which:

$$m_j(u, \mathbf{z}) = \sum_{r=1}^R b_{rj} u^r + \sum_{t=1}^T g_{tj} z_t, \quad (7)$$

and,

$$a_{jk}(\mathbf{z}) = a_{jk} \quad \text{where} \quad a_{jk} = a_{kj} \quad \forall j \neq k \quad (8)$$

Substitute equations Eq. 3, 7 and 8 into Eq. 6, we can derive the formula of budget share accommodating energy demand, EASI-E, as follows,

$$w_j = \sum_{r=1}^R b_{rj} u^r + \sum_{t=1}^T g_{tj} z_t + \sum_{k=1}^J a_{jk} \ln p_k + \sum_{k=1}^E \phi_{jk}(e_j)^{\gamma_j} (e_k)^{\gamma_k} u \ln p_k + \varepsilon_j. \quad (9)$$

In addition, from Eq. 6, we can take out $m_j(u, \mathbf{z})$ to the left-hand side, multiply with the corresponding price $\ln p_k$, then accumulate all expressions of all goods j together to have:

$$\begin{aligned} \sum_{j=1}^J m_j(u, \mathbf{z}) \ln p_j &= \sum_{j=1}^J w_j \ln p_j - \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k \\ &\quad - \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_j \ln p_k - \sum_{j=1}^J \varepsilon_j \ln p_j. \end{aligned} \quad (10)$$

Substituting $\sum_{j=1}^J m_j(u, \mathbf{z}) \ln p_j$ from Eq. 10 into Eq. 4 and replacing the term $C(\mathbf{p}, u, \mathbf{z}, \mathbf{e}, \gamma, \varepsilon)$ by observed total expenditure x , we can transform the Eq. 4 to Eq. 11,

$$\ln x = u + \sum_{j=1}^J w_j \ln p_j - \frac{1}{2} \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k - \frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) u \ln p_j \ln p_k. \quad (11)$$

Then, we can derive the implied implicit utility function u by the following formula:

$$u = \frac{\ln x - \sum_{j=1}^J w_j \ln p_j + \frac{1}{2} \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k}{1 - \frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) \ln p_j \ln p_k} \quad (12)$$

In summary, the EASI-E demand system will simultaneously estimate the system of j equations given in Eq. 9 with an endogenous implied utility provided in Eq. 12, i.e., the finalized EASI-E demand system can be written as,

$$\begin{aligned} w_j &= \sum_{r=1}^R b_{rj} \left(\frac{\ln x - \sum_{j=1}^J w_j \ln p_j + \frac{1}{2} \sum_{j=1}^J \sum_{k=1}^J a_{jk}(\mathbf{z}) \ln p_j \ln p_k}{1 - \frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) \ln p_j \ln p_k} \right)^r \\ &\quad + \sum_{t=1}^T g_{tj} z_t + \sum_{k=1}^J a_{jk} \ln p_k + \sum_{k=1}^E \phi_{jk} e_j \exp(\gamma_j) e_k \exp(\gamma_k) u \ln p_k + \varepsilon_j \end{aligned} \quad (13)$$

We already defined a household as in energy poverty if the implied utility is lower than the mean of all households' utilities. Thus, energy poverty exists *iff*

$$\Delta u_i = u_i - \bar{u} < 0, \quad (14)$$

where u_i denotes the utility of i^{th} household that will be calculated in Eq. 12 by the EASI-E estimations. For simplicity, the mean of all households' utilities, $\bar{u} = \frac{\sum_{i=1}^n u_i}{n}$, is determined by the average implied utility of all households in the country (where n is the total number of households).³

The EASI-E system will collapse to the original EASI in the absence of trade-offs between indoor emissions and energy prices. This will occur if

- (i) the price of alternative energy drops to zero, i.e., $\ln p_k = 0$, which happens if that alternative energy is no longer available; or
- (ii) the household uses only one fuel, i.e., $e_j = 1$ and $e_k = 0 \forall k \neq 0$, which happens if one energy is dominantly preferred or the only one available. In these two cases: the term $\frac{1}{2} \sum_{j=1}^E \sum_{k=1}^E \phi_{jk}(\mathbf{e}, \gamma) \ln p_k$ in Eq. 12 and the implied utility modifier PEP in Eq. 13, i.e., $PEP = \sum_{k=1}^E \phi_{jk} e_j \exp(\gamma_j) e_k \exp(\gamma_k) u \ln p_k$, will be equal to zero to reflect a household's decision to stick with the only option.
- (iii) all energies have zero indoor emission, i.e., $\gamma_j = \gamma_k = 0$, households will no longer compare energies based on their cleanness. Instead, households' consideration of energy mix becomes naturally price compared, and the Eq. 12 and Eq. 13 will collapse to the EASI model having interaction term between price and utility, which was discussed in Pendakur (2009).

Like the original EASI demand system, the EASI-E system is non-linear and endogenous. The EASI-E system can be estimated using a generalized method of moments (GMM) or an iterated linear estimator with similar performance. In this study, we choose the iterated linear estimator to avoid potential overidentified GMM models in the case of small samples, as suggested in Pendakur (2009). To verify the performance of EASI-E in identifying energy poverty, we will compare the result with the two alternative models, (i) IEA definition of energy poverty and (ii) LIHC proposed by Hills (2011).

4. DATA AND ESTIMATION

4.1 Data

Household data

We applied household data from the Vietnam Household Living Standards Survey (VHLSS), conducted every two years by the General Statistics Office of Vietnam (GSOV). The survey is designed according to international standards to systematically monitor the living standards of Vietnamese people (Nguyen et al., 2019). It also helps assess government strategies and development goals, such as the Comprehensive Poverty Reduction and Growth Strategy, the Millennium Development Goals and the Sustainable Development Goals. In addition, the VHLSS data provides detailed household income, expenditure, composition, demographics, housing characteristics, and geographical information. Although the VHLSS data is not publicly accessible, it can be acquired from GSOV upon

3. The mean of all households' utilities, $\bar{u}$, could be flexibly calculated in different ways to suit various study purposes. It could be the national average or regional average, depending on the geographic scope of the study. In addition, $\bar{u}$ could be calculated as the mean of ordinary utilities, which do not consider the implied emission impacts of energy use. That calculation is appropriate for a study focusing on how divergent a household's utility could be, once they acknowledge the emission effect. It is, however, required to have an addition estimation by the original EASI model. Finally, $\bar{u}$ could be calculated as the mean of households' implied utilities, which is applied by this study. This approach is simpler because no additional estimation is required. It is also more reasonable in the assessment of energy poverty because it references the energy consumption choices of all households.

request and has been widely used in numerous academic research, such as McCaig and Pavcnik (2018, 2015) and Nguyen et al. (2019).

Although possessing all datasets of VHLSS from 2002 to 2016, we will only adopt three surveys with approximately 24,300 observations in the year 2012, 2014 and 2016 because the regional price data required to conduct the model is only available from 2012. In order to prepare data for the model, we categorized non-durable, non-energy annual expenditures into 11 categories that are consistent with the classification of GSOV. Those categories are food, tobacco and drink, clothing and footwear, housing costs, house goods, medication, transportation, communication, education, leisure, and others. In addition, we extracted the energy expenditures used for living purposes but not for transportation and categorized them into three groups, (a) electricity, (b) LPG, and (c) conventional cooking energy types. The major purpose was to group energy types based on (a) their emission levels, (b) the purpose and equipment used, and (c) the availability of energy prices data, which will be discussed in the next part.

Price data

The regional price indices are available from 2012 for all six regions of Vietnam, namely, the Red River Delta, Northern Midland and Mountains, North Central and Central Coast, Central Highlands, the Southeast, and the Mekong River Delta.⁴ Due to the difficulties in collecting the specific prices of conventional cooking energy types (e.g., peat briquette, low-quality kerosene, wood chips, or plant biomass), price indices of all such categories are provided in a composite index.

Following Lewbel and Pendakur (2009), we use the normalized prices, wherein the price vector facing the Red River Delta area in 2012 is $(1, \dots, 1)$, which is also compatible with the data provided by the GSOV. In addition, although the normalized prices are not actual monetary prices, they can reflect precisely the evolution of the price in each category and the diversity of the price across regions (Tovar Reaños and Wölfling, 2018).

Emission rate

Emissions of interest are those related to energy use's indoor pollution, which causes many household health problems. These emissions may include smoke, dust, fly ash, and greenhouse gases, such as CO_2 , SO_x or NO_x . However, except for greenhouse gases, the remaining indicators are not available to be collected or calculated at the household level. Therefore, we only use the greenhouse gas emission index, converted to CO_2 equivalent measurement, to represent all indoor emissions in this study. Considering that, the emission rates of non-energy goods are zero because consuming all non-energy goods in a household does not produce any indoor pollution. Among three types of energy, electricity is considered the cleanest energy for household use (Nguyen et al., 2019). In fact, household electricity use, including for all the purposes such as cooking, lighting, or running electrical appliances, does not show any perceived indoor emission to the users. Thus, we can assume that the CO_2 emission rate of electricity used in a household is zero for analyzing consumer behavior concerning energy emissions.

Meanwhile, the emissions of the other two energy groups, which are LPG and conventional cooking energy types, are more readily observable. The emission rates of those energies are collected from the default emission factors for stationary combustion in the residential, established in the Guidelines for National Greenhouse Gas Inventories by the Intergovernmental Panel on Climate Change (IPCC) (Eggleston et al., 2006). We adopt the composite emission rate for conventional

4. All regional prices for non-energy goods are publicly published on the GSOV website, while energy prices according to three energy groups mentioned above are only available upon request.

cooking energies, calculated as the average emission rate of the most common cooking fuels in Vietnam, i.e., low-quality kerosene, brown coal briquettes, charcoal, or wood waste. The emission rates are measured in terms of a ton of CO_2 per kiloton of oil equivalent (ton CO_2 /ktoe) to facilitate the estimation.

Table 1 presents the descriptive statistics of these variables.

Table 1: Variable Descriptive Statistics

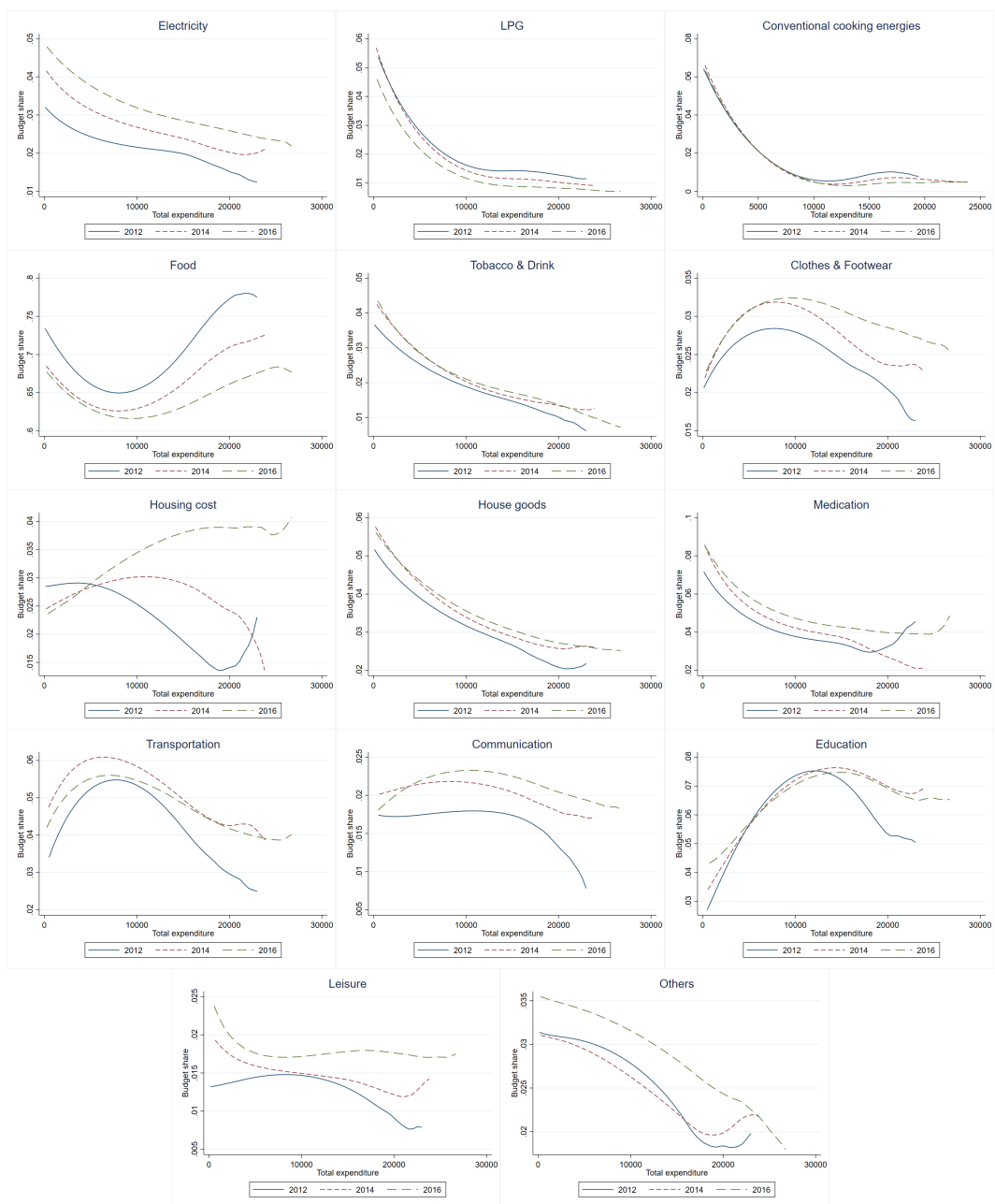
	Variable	Obs	Mean	Std. Dev.	Min	Max
Budget share	Electricity	24,297	0.030	0.021	0.000	0.357
	LPG	24,297	0.015	0.016	0.000	0.276
	Conventional cooking energy types	24,297	0.013	0.021	0.000	0.579
	Food	24,297	0.648	0.113	0.031	0.954
	Tobacco & Drink	24,297	0.017	0.023	0.000	0.350
	Clothes & Footwear	24,297	0.029	0.020	0.000	0.206
	Housing cost	24,297	0.017	0.056	0.000	0.844
	House goods	24,297	0.041	0.020	0.000	0.288
	Medication	24,297	0.050	0.077	0.000	0.922
	Transportation	24,297	0.050	0.040	0.000	0.435
	Communication	24,297	0.018	0.016	0.000	0.145
	Education	24,297	0.036	0.055	0.000	0.681
	Leisure	24,297	0.008	0.016	0.000	0.435
	Others	24,297	0.029	0.027	0.000	0.473
Price index (log-form)	Electricity	24,297	0.025	0.060	-0.073	0.142
	LPG	24,297	0.027	0.021	-0.010	0.068
	Coal and solid biomass	24,297	-0.125	0.181	-0.398	0.020
	Food	24,297	-0.001	0.012	-0.023	0.018
	Tobacco & Drink	24,297	0.010	0.008	0.000	0.026
	Clothes & Footwear	24,297	0.005	0.015	-0.026	0.024
	Housing cost	24,297	0.010	0.014	-0.008	0.047
	House goods	24,297	0.004	0.011	-0.017	0.022
	Medication	24,297	0.052	0.052	-0.020	0.171
	Transportation	24,297	-0.005	0.012	-0.019	0.021
	Communication	24,297	0.046	0.030	0.000	0.078
	Education	24,297	0.013	0.038	-0.035	0.136
	Leisure	24,297	0.001	0.011	-0.020	0.016
	Others	24,297	0.013	0.012	-0.006	0.040
Demographics	Urban/rural	24,297	1.726	0.446	1.000	2.000
	Household size	24,297	3.737	1.537	1.000	12.000
	Highest education in household	24,016	2.876	2.507	0.000	12.000
Financial	Non-durable expenditure	24,297	8.538	0.662	4.369	10.193
	Total income	24,297	8.700	0.799	4.774	12.855

4.2 Estimation

As a preliminary analysis, we estimate some simple non-parametric regressions depicting Engel curves for the data set. Figure 4 shows that the Engel curves representing most goods are non-linear, while electricity, tobacco & drinks, and house goods exhibit only slight non-linearity. Given that the estimated Engel curves are non-linear and could be of high-order polynomials, the EASI-E demand model is appropriate for this exercise (Lewbel and Pendakur, 2009).

The econometric analysis involves the estimation of Eq. 13. In this estimation, we only implement the EASI-E demand system with a polynomial of order three in u . We tested with higher-order polynomials up to an order of five, but they were not significant and reduced the significance level of other variables in the model. The EASI-E demand system of 14 goods is estimated and presented in Table 2. All components of the implied emission expenditure share

Figure 4: Non-parametric Engel Curves of Expenditure Share of Non-durable Goods vs Total Expenditure



Source: The authors conducted this figure with the actual data from the dataset using Stata software.

Notes: straight line: year 2012; dash line: year 2014; dot line: year 2016.

$(EP_{j,k} = \phi_{jk}(\mathbf{e}, \gamma) u \ln p_k)$ in energy-budget share equations are highly significant. This suggests that the emissions associated with the energy mix of households significantly impact their energy consumption behavior. It further demonstrates that the model we propose is valid in the context of consumer behavior, i.e., consumers always face the implied utility modifier (PEP) when energy sources are associated with emissions.

Table 2: Estimation Output of EASI-E Demand System

	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	w_9	w_{10}	w_{11}	w_{12}	w_{13}	w_{14}
$EP_{1,2}$	0.0015*** (0.000)													
$EP_{1,3}$	0.0001*** (0.002)													
$EP_{2,1}$	0.0015*** (0.000)													
$EP_{2,3}$	-0.0001*** (0.000)													
$EP_{3,1}$			0.0001*** (0.002)											
$EP_{3,2}$			-0.0001*** (0.000)											
$\ln p_1$	0.1380*** (0.000)	0.0029 (0.777)	0.0151*** (0.000)	-0.1536*** (0.000)	-0.1479*** (0.000)	-0.0659*** (0.002)	0.0906*** (0.000)	0.0569** (0.015)	0.0237*** (0.004)	-0.0391** (0.029)	0.0125** (0.042)	0.0268*** (0.003)	0.0245 (0.130)	-0.0240 (0.134)
$\ln p_2$	0.0029 (0.777)	-0.1740*** (0.000)	0.0471*** (0.000)	0.1886*** (0.001)	0.0813* (0.080)	0.1193*** (0.008)	-0.1823*** (0.000)	-0.0176 (0.687)	0.0427*** (0.000)	-0.2453*** (0.000)	0.0579*** (0.000)	-0.0058 (0.539)	0.08023** (0.013)	0.0498 (0.247)
$\ln p_3$	0.0151*** (0.000)	0.0471*** (0.000)	-0.0106*** (0.000)	-0.0283*** (0.047)	-0.0505*** (0.000)	-0.0195** (0.032)	0.0307*** (0.000)	0.0160* (0.083)	-0.0148*** (0.003)	0.0164** (0.035)	-0.0014 (0.603)	0.0259*** (0.000)	-0.0007 (0.920)	-0.0351*** (0.000)
$\ln p_4$	-0.1536*** (0.000)	0.1886*** (0.001)	-0.0283** (0.047)	2.3720*** (0.000)	-0.4831*** (0.001)	-0.3609** (0.016)	-1.0759*** (0.000)	-0.2087 (0.129)	0.0622* (0.096)	-0.1240 (0.322)	0.3573*** (0.000)	-0.1115*** (0.003)	-0.1319 (0.213)	-0.7919*** (0.000)
$\ln p_5$	-0.1479*** (0.000)	0.0813* (0.080)	-0.0505*** (0.000)	-0.4831*** (0.001)	-0.5253*** (0.006)	-0.0336 (0.826)	0.3303*** (0.001)	0.0980 (0.572)	-0.0988*** (0.000)	0.0188 (0.834)	0.0583** (0.028)	0.0336*** (0.007)	0.5117*** (0.000)	0.0288 (0.760)
$\ln p_6$	-0.0659*** (0.002)	0.1193*** (0.000)	-0.0195** (0.032)	-0.3609*** (0.016)	-0.0336 (0.826)	0.1991 (0.323)	0.1082 (0.165)	-0.1218 (0.514)	-0.0071 (0.691)	0.1764* (0.073)	-0.1301*** (0.000)	-0.0056 (0.607)	-0.0723 (0.543)	0.2961*** (0.001)
$\ln p_7$	0.0906*** (0.000)	-0.1823*** (0.000)	0.0307*** (0.000)	-1.0759*** (0.000)	0.3303*** (0.001)	0.1082 (0.165)	0.0463 (0.611)	0.0718 (0.386)	0.0683*** (0.001)	0.4521*** (0.000)	-0.1285*** (0.000)	0.0415* (0.063)	0.2083*** (0.000)	0.1536** (0.046)
$\ln p_8$	0.0569** (0.015)	-0.0176 (0.687)	0.0160* (0.083)	-0.2087 (0.129)	0.0980 (0.572)	-0.1218 (0.514)	0.0718 (0.386)	0.4392 (0.100)	0.0173 (0.329)	-0.0174 (0.845)	-0.0590** (0.041)	-0.0054 (0.041)	-0.3877*** (0.607)	0.1636* (0.002)
$\ln p_9$	0.0237*** (0.004)	0.0427*** (0.000)	-0.0148*** (0.000)	0.0622** (0.096)	-0.0988*** (0.000)	-0.0071 (0.691)	0.0683*** (0.001)	0.0173 (0.329)	-0.0847*** (0.000)	-0.0740*** (0.000)	0.0379*** (0.000)	0.0621*** (0.000)	0.0286* (0.050)	-0.0632*** (0.000)
$\ln p_{10}$	-0.0391** (0.029)	-0.2453*** (0.000)	0.0164** (0.035)	-0.1240 (0.322)	0.0188 (0.834)	0.1764* (0.073)	0.4521*** (0.000)	-0.0174 (0.845)	-0.0740*** (0.000)	-0.5538*** (0.000)	-0.0146 (0.447)	0.0473*** (0.004)	0.2684*** (0.000)	0.3974*** (0.000)
$\ln p_{11}$	0.0125*** (0.042)	0.0579*** (0.000)	-0.0014 (0.603)	0.3573*** (0.000)	0.0583*** (0.028)	-0.1301*** (0.000)	-0.1285*** (0.000)	-0.0590** (0.041)	0.0379*** (0.000)	-0.0146 (0.447)	-0.0894*** (0.000)	-0.0343*** (0.000)	-0.1368*** (0.000)	0.0353** (0.081)
$\ln p_{12}$	0.0268*** (0.003)	-0.0058 (0.539)	0.0259*** (0.000)	-0.1115*** (0.003)	0.0336*** (0.007)	-0.0056 (0.607)	0.0415* (0.063)	-0.0054 (0.607)	0.0621*** (0.000)	0.0473*** (0.004)	-0.0343*** (0.000)	-0.0168 (0.368)	-0.0062 (0.453)	-0.0051 (0.693)
$\ln p_{13}$	0.0245 (0.130)	0.0802** (0.013)	-0.0007 (0.920)	-0.1319 (0.213)	0.5117*** (0.000)	-0.0723 (0.543)	0.2083*** (0.000)	-0.3877*** (0.002)	0.0286* (0.050)	0.2684*** (0.000)	-0.1368*** (0.000)	-0.0062 (0.453)	-0.6636*** (0.000)	0.1403** (0.042)
$\ln p_{14}$	-0.0240 (0.134)	0.0498 (0.247)	-0.0351*** (0.000)	-0.7919*** (0.000)	0.0288 (0.760)	0.2961*** (0.001)	0.1536** (0.046)	0.1636* (0.067)	-0.0632*** (0.000)	0.3974*** (0.000)	0.0353** (0.081)	-0.0051 (0.693)	-0.1403** (0.042)	-0.2554** (0.017)
Urban/rural	-0.0054*** (0.000)	0.0101*** (0.000)	-0.0174*** (0.000)	-0.0146*** (0.000)	0.0013 (0.155)	0.0046*** (0.000)	-0.0034* (0.064)	0.0021*** (0.006)	0.0106*** (0.000)	0.0050*** (0.000)	0.0010 (0.207)	0.0046** (0.019)	-0.0004 (0.503)	0.0001 (0.886)
HH size	-0.0016*** (0.000)	-0.0113*** (0.000)	0.0091*** (0.000)	0.0190*** (0.000)	0.0009*** (0.003)	0.0006** (0.012)	-0.0060*** (0.000)	-0.0012*** (0.000)	-0.0046*** (0.000)	-0.0037*** (0.000)	-0.0026*** (0.000)	0.0056*** (0.000)	-0.0014*** (0.000)	-0.0008*** (0.007)
Education	0.0011*** (0.000)	-0.0029*** (0.000)	0.0038*** (0.000)	-0.0009 (0.148)	-0.0007*** (0.000)	0.0000 (0.855)	-0.0017*** (0.000)	0.0002 (0.294)	-0.0008** (0.043)	0.0011*** (0.000)	0.0006*** (0.000)	0.0018*** (0.000)	0.0003*** (0.006)	-0.0011*** (0.000)
$\ln y$	-0.0240 (0.489)	-0.2013*** (0.000)	-0.3370*** (0.000)	0.7091*** (0.000)	0.9373 (0.213)	2.0852*** (0.001)	1.3014 (0.359)	0.3096 (0.618)	0.1540** (0.021)	2.8509*** (0.005)	-2.8300*** (0.000)	-3.2449*** (0.024)	0.2421 (0.629)	-0.1611 (0.839)
$(\ln y)^2$	0.0082 (0.325)	0.0444*** (0.000)	0.0876*** (0.000)	-0.1348*** (0.000)	-0.1114 (0.231)	-0.2633*** (0.001)	-0.1672 (0.340)	-0.0311 (0.686)	-0.0338** (0.035)	-0.3035*** (0.015)	0.3395*** (0.000)	0.3582** (0.044)	-0.0314 (0.612)	0.0178 (0.857)
$(\ln y)^3$	-0.0006 (0.255)	-0.0024*** (0.000)	-0.0056*** (0.000)	0.0070*** (0.000)	0.0044 (0.251)	0.0110*** (0.001)	0.0072 (0.318)	0.0010 (0.756)	0.0019** (0.044)	0.0106** (0.038)	-0.0134*** (0.000)	-0.0130* (0.076)	0.0014 (0.589)	-0.0006 (0.878)
Constant					-2.5963 (0.199)	-5.4515*** (0.001)	-3.3324 (0.381)	-0.9467 (0.571)		-8.7771*** (0.001)	7.8015*** (0.000)	9.6460** (0.013)	-0.6230 (0.643)	0.4944 (0.817)
Observations	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016	24,016

p -values in parentheses: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$;

Notes:

* $w_1, w_2, \dots, w_{14}$ and $\ln p_1, \ln p_2, \dots, \ln p_{14}$ are respectively budget shares and log of prices of (1) Electricity, (2) LPG, (3) Conventional cooking energies, (4) Food, (5) Tobacco & Drink, (6) Clothes & Footwear, (7) Housing cost, (8) House goods, (9) Medications, (10) Transportation, (11) Communication, (12) Education, (13) Leisure, (14) Others.

* terms $EP_{jk} = \varphi_{jk}(e_j)^{\gamma_j}(e_k)^{\gamma_k} u \ln p_k, \forall j \neq k \text{ and } j, k \in \{1, 2, 3\}$, reflects the substitution effect caused by emissions which is explained in Eq. 3. Thus, EP_{jk} appear only in equations of energies, include (1) Electricity, (2) LPG, (3) Conventional cooking energies.

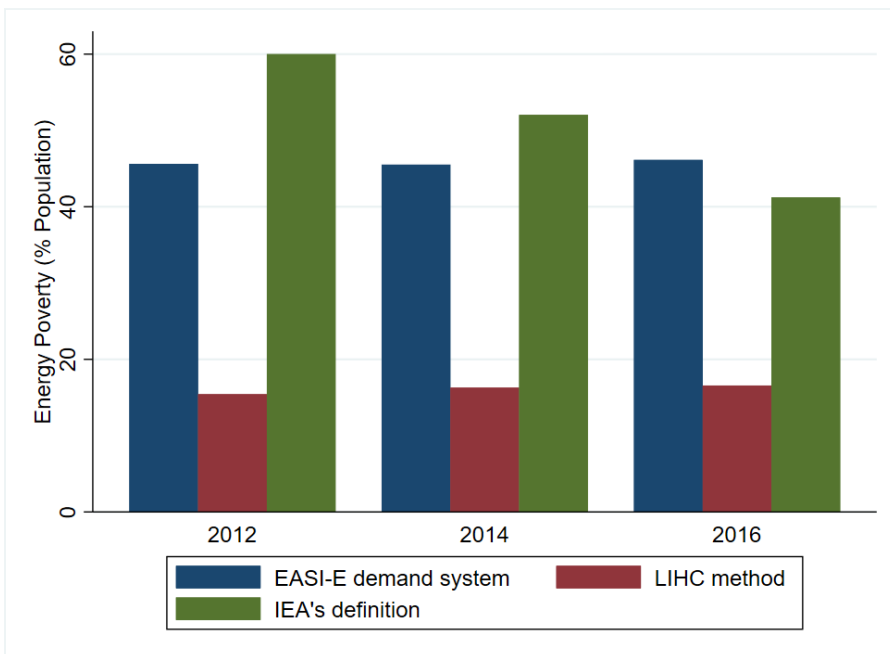
5. IMPLICATIONS AND FINDINGS

As discussed above, we can compare a specific household's implied utility, u_i , with the national average utility, $\bar{u} = \frac{\sum_{i=1}^n u_i}{n}$, to determine if the household is energy-poor. We have estimated the proportion of energy poverty households in Vietnam for three different survey periods (2012, 2014, &

2016). In order to compare our estimations with traditional energy poverty measures, we calculated energy poverty using (a) low-income high cost and (b) the percentage of people using conventional cooking fuels that resembles the IEA definition. Figure 5 shows the results of the three methods.

According to Figure 5, our results are distinct from both IEA estimates and the LIHC method. The result using the IEA's definition shows that the percentage of households living in energy poverty declined consistently from 2012 to 2016 because the measure of energy poverty relied completely on the use of conventional fuels. Therefore, this definition cannot distinguish households consuming small amounts of polluted energies while mainly using modern energies from the ones that do not have any financial constraints on switching to modern energies completely. Meanwhile, the number of energy poverty households based on LIHC measurement slightly and consistently increased from 2012 to 2016. This result may explain the case when income and energy costs were both increasing during that period. However, it is obviously observed that numerous middle-income households choose conventional fuels over modern energies to save money for other demands. Also, many wealthy households in rural areas continue to consume large amounts of traditional energy habitually. The use of cheap fuels reduces those households' energy expenditure share, resulting in them being overlooked by the LIHC. However, many of them may be considered energy poverty because their family's health is at high risk.

Figure 5: Different Results from Proposed EASI-E system vs. LIHC and IEA Methods



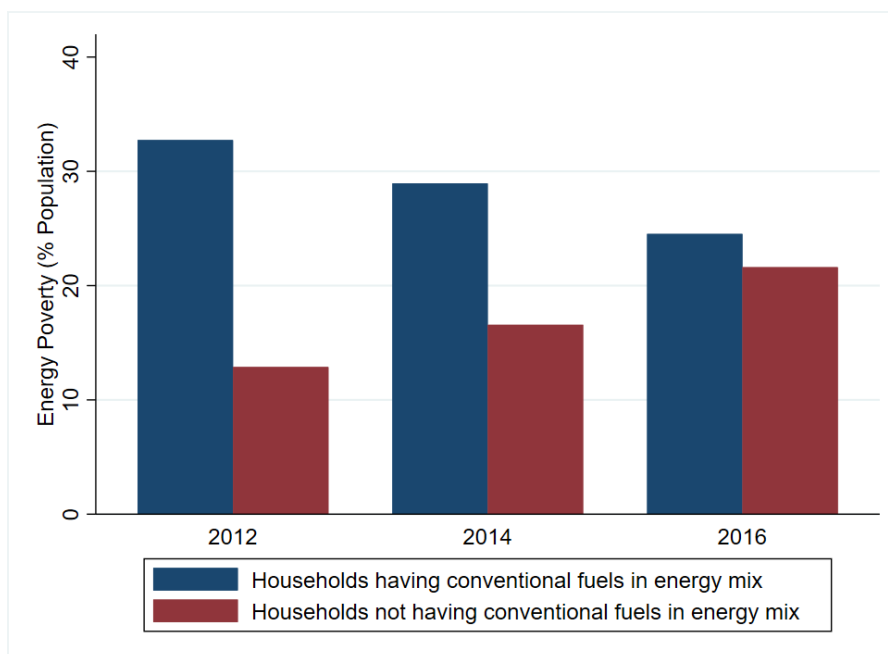
Source: The authors conducted this figure with the estimation outputs by three methods using Stata software.

Results from the EASI-E demand system are different from the aforementioned methods. The number of energy poverty households reduced slightly in 2014, before slightly increasing in 2016. The EASI-E demand system includes all potential terms such as households' income, regional prices of energy and other goods, and households' awareness of emissions and health risks to determine energy poverty via implied disutility. Thus, the different trend of EASI-E compared to other methods implies a higher capability in capturing both the impact of emissions associated with con-

ventional fuels and the impact caused by movements in prices. We will discuss this in more detail in the following figure.

In Figure 6, we attempt to explore further the EASI-E capability in poverty estimates. We disaggregated energy poverty into two groups. Group 1 consists of households with conventional energy in their energy mix, and Group 2 is completely conventional energy free. As illustrated in Figure 6, approximately 13% of the households were in energy poverty while not using traditional fuel in 2012, and that number in 2016 was 21.6%. Those percentages implied that disutility could be derived from the trade-off of many other demands for clean energy, not only from the pollution associated with traditional fuels.

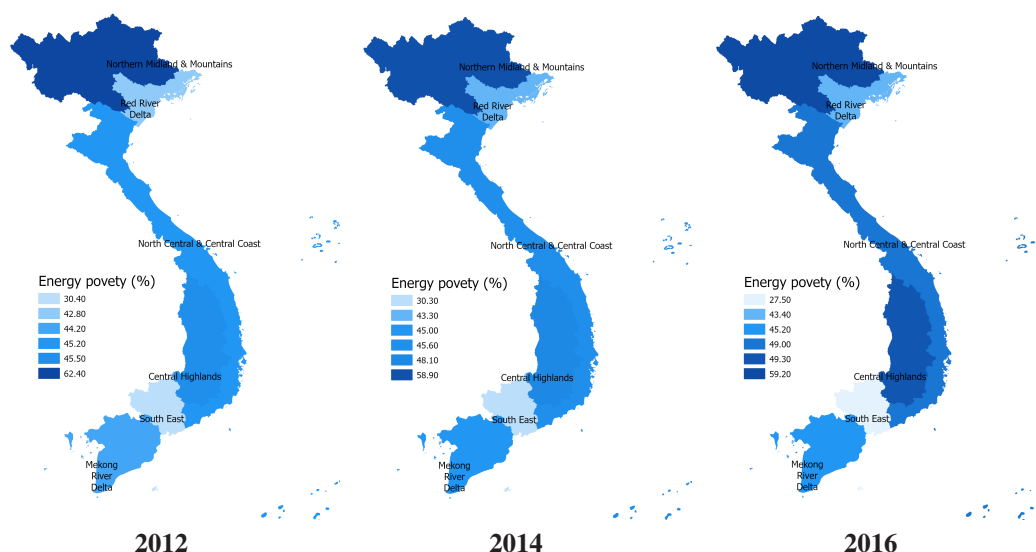
Figure 6: Energy Poverty Distribution by the Conventional Fuels Use (or Not) over the Years



Source: The authors conducted this figure with the estimation outputs by three methods using Stata software.

Furthermore, according to our estimates, energy poverty in Group 1 decreased as expected, implying that households are gradually moving away from conventional fuels, consistent with IEA prediction. Meanwhile, the poverty in Group 2 increased from 2012 to 2016. These observations are explained by the changes in both income and energy prices. The increase in household living standards and urbanization contributed to reducing the use of polluting energy in households over the period. However, the rise in energy prices, which was more dramatic in modern energies, was putting higher pressure on households' budgets and led to more severe disutility among households.

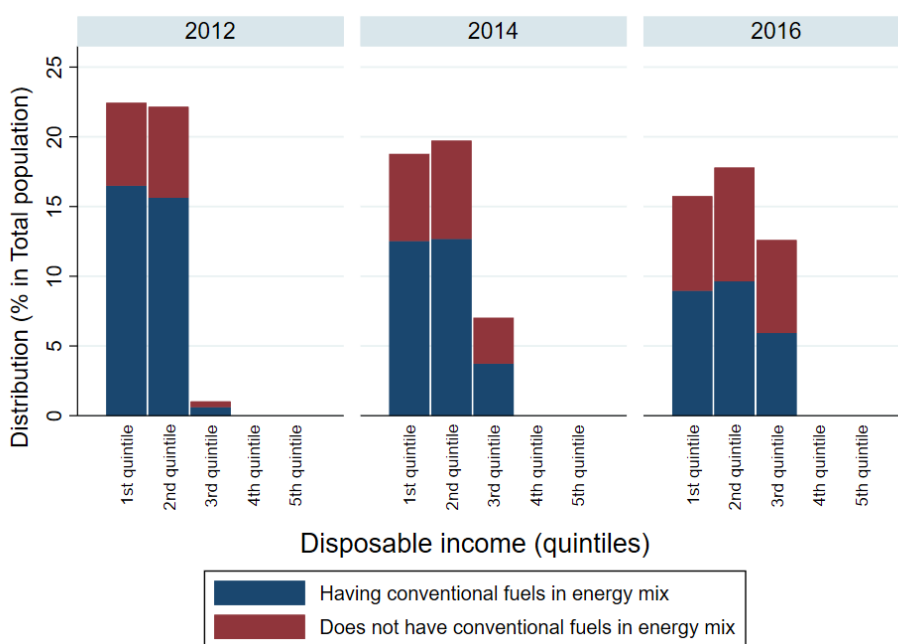
In Figure 7, we illustrate the regional differences between EASI-E energy poverty estimates from 2012 to 2016. As can be seen from the graph, the Southeast region, where the finance capital (Ho Chi Minh city) is located, had the lowest records for energy poverty in the whole period. In opposite, the most rural and highland areas, i.e. Northern Midland & Mountains, had the worst energy poverty. In addition, except the Southeast region gradually declined in energy poverty, other regions showed a decline followed by an increase in 2016 or a steady increase in the whole period. It reflects that the change in energy price has a limited impact on a wealthy region but has significant adverse impacts on other regions, where the majority of the population is rural and poor.

Figure 7: Energy Poverty Distribution by the Region over the Years

Source: The authors conducted this figure with the estimation outputs by the EASI-E method using Stata software and illustrated the results on Vietnam's map by ArcGIS software.

In the next exercise, we explore the distribution of energy poverty incidents across different income groups. In contrast to the LIHC method, we believe that low household income levels are not necessarily a significant energy poverty indicator. Instead, households at higher income levels could be in energy poverty, and it could be because of excessive use of dirty energies in the energy mix or scarifying other demands to consume clean energy. Figure 8 shows the distribution of energy poverty in households at different income quintiles, categorized by the energy types and separated for the three survey periods.

Figure 8 provides three insights. First, although most energy poverty households fell into lower-income groups, a remarkable increasing trend is that homes in higher income levels had also fallen into energy poverty. For example, approximately 12% of households in energy poverty were in the third quintile of disposable income in 2016, increasing by about six times compared to 2012. The trend implies that an income level that was less likely to be energy poor could become energy poverty in the future. Furthermore, that is the basis to argue against choosing only low income to assess energy poverty in existing methods. Second, Figure 8 demonstrates that the conventional fuel use dramatically decreased in households in the first two quintiles but significantly increased in the third quintile, especially when the energy price rose. Thus, it implies that income or wealth level is not necessarily a motivation for families to climb the energy ladder. Third, although certain households in high-income groups consumed traditional biomass, as shown in Figure 4, there were no homes from wealthy cohorts in energy poverty, as in Figure 8. Thus, we could state that using traditional biomass does not necessarily link to energy poverty, as long as households do not have the disutility. However, from the previous discussion, it is still potential that those families may fall into energy poverty in the future.

Figure 8: Energy Poverty Distribution by Income Levels over the Years

Source: The authors conducted this figure with the estimation outputs by three methods using Stata software.

Emerging requirements of policies on energy poverty

In most research and policy works, energy poverty is virtually synonymous with income poverty. For instance, accessibility methods focus on families relying on traditional biomass and not having access to modern energy services, and affordability methods like the LIHC only consider households below 60% of the median income level. In addition, many countries even do not have an appropriate measure of energy poverty and, therefore, no clear policies to tackle the energy poor. For example, Vietnam does not have an official definition of energy poverty. Instead, the Vietnamese government promulgated a policy to monthly subsidize 30 kWh of electricity for income-poor households with electricity usage below 50kWh/month. ⁵

Conversely, our study finds that the ties in the definition of energy poverty associated with income poverty or low use of modern energy are unreasonable and inadvertently create a barrier to society's progress. Both low-income and middle-income households are vulnerable if they struggle to afford clean energy or risk their health by consuming too much polluting fuel. Therefore, a good energy poverty policy should support both and provide incentives, e.g. financial support or a special price scheme, to consume more clean energy. In addition, high-income level households who are not (or not yet) in energy poverty but are overconsuming carbon-intensive fuels also need to be addressed. These households do not need any support program as they can easily switch their energy use; however, they need to be alerted that their energy use creates many health risks.

6. CONCLUSION

Current approaches to energy poverty measurement invoke some strong and sometimes implausible assumptions. They can presume energy-poor occurs if and only if household income

is below a certain threshold. In addition, they expect households to progress through the energy ladder as income increases to move away from polluted traditional fuels. However, data show that even in developed nations, the energy mix of households could consist of both carbon-intensive traditional biomass and clean, modern energy. Therefore, the current energy poverty measures, mainly based on affordability and accessibility to modern energy, hold limited relevancy, especially in countries with a varied residential energy mix. To this end, we have developed a new energy poverty measure that can cover the gaps in existing methods to overcome their drawbacks and involve energy consumption behavior and emission impacts.

The results from our EASI-E demand system are different from conventional methods because we have included implied disutility, either due to indoor emissions and potential health risks or sacrificing other demands to use clean energy. Therefore, the energy poverty estimates from this method are higher than conventional methods.

According to the results, the number of households living in energy poverty in Vietnam remained around 44% from 2012 to 2016. The findings also show that household income may not lead households to climb the energy ladder because many families in the highest wealth level continue to consume conventional fuels. In addition, homes at different income levels could be in energy poverty if they experience disutility. However, we also find that consuming carbon-intensive traditional biomass does not necessarily mean being energy-poor if the implied utility is not lower than the national average level.

From these findings, we suggest that the definition of energy poverty should not be tied to the income-poor but be expanded, where households at any income level could be in energy poverty. In terms of policy, low-income and middle-income households in energy poverty could be targeted for incentives, e.g. financial support or a special price scheme. In addition, higher-income families who consume dirty fuels should be alerted to the potential health risks associated with this choice.

Some limitations exist in this paper that could potentially be relevant for further studies. First, in Vietnam, the choice of clean energy over traditional energy always creates a negative income effect and vice versa because the prices of traditional energies are always much lower than the prices of modern energies. If this method is applied to other countries, careful consideration must be paid to prices as well as energy efficiency.

Second, we used the energy composite price index and energy expenditure as a proxy for calculating the implied utility because actual price and energy consumption data were unavailable. Access to actual energy consumption data could improve our modeling approach.

Third, detailed prices of goods, especially energy prices, play a significant role in estimating the demand system and evaluating households' utility. The use of the regional average price index reduces the heterogeneity across households, although it is acceptable, as suggested by Lewbel and Pendakur (2009). More detailed price data at the household level could increase the significance of the model.

Finally, although the EASI-E system offers promise in modeling energy poverty, it requires a well-structured and fully-fledged dataset, which might not be accessible in many developing countries.

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REFERENCES

- Abbas, K., S. Li, D. Xu, K. Baz, and A. Rakhmetova (2020). “Do socioeconomic factors determine household multidimensional energy poverty? Empirical evidence from South Asia.” *Energy Policy* 146. ISSN 03014215. <https://doi.org/10.1016/J.ENPOL.2020.111754>.
- Action, P. (2010). *Poor people's energy outlook 2010*. Practical Action Publishing. ISBN 1780447736.
- Barnes, D. F., S. R. Khandker, and H. A. Samad (2011). “Energy poverty in rural Bangladesh.” *Energy Policy* 39(2): 894–904. <https://doi.org/10.1016/j.enpol.2010.11.014>.
- Bhide, A. and C. R. Monroy (2011). “Energy poverty: A special focus on energy poverty in India and renewable energy technologies.” *Renewable and Sustainable Energy Reviews* 15(2): 1057–1066. <https://doi.org/10.1016/j.rser.2010.11.044>.
- Biröl, F. (2007). “Energy economics: a place for energy poverty in the agenda?” *The Energy Journal* 28(3). Boardman, B. (1991). *Fuel poverty: from cold homes to affordable warmth* Pinter Pub Limited. ISBN 1852931396.
- Boardman, B. (2012). “Fuel poverty synthesis: Lessons learnt, actions needed.” *Energy Policy* 49: 143–148. <https://doi.org/10.1016/j.enpol.2012.02.035>.
- Charlier, D. and S. Kahouli (2019). “From Residential Energy Demand to Fuel Poverty: Income-induced Non-linearities in the Reactions of Households to Energy Price Fluctuations” *The Energy Journal* 40(2): 101–137. ISSN 1944-9089. <https://doi.org/10.5547/01956574.40.2.DCHA>.
- Charlier, D. and B. Legendre (2019). “A Multidimensional Approach to Measuring Fuel Poverty.” *The Energy Journal* 40(2): 27–53. ISSN 1944-9089. <https://doi.org/10.5547/01956574.40.2.BLEG>.
- Delugas, E. and R. Brau (2021). “Evaluating the Impact of Energy Poverty in a Multidimensional Setting.” *The Energy Journal* 42(1). ISSN 1944-9089. <https://doi.org/10.5547/01956574.42.1.EDEL>.
- Desa, U. N. (2016). “Transforming our world: The 2030 agenda for sustainable development.”
- Dietz, S. and F. Venmans (2019). “Cumulative carbon emissions and economic policy: in search of general principles.” *Journal of Environmental Economics and Management* 96: 108–129.
- Dogan, E., M. Madaleno, R. Inglesi-Lotz, and D. Taskin (2022). “Race and energy poverty: Evidence from African-American households.” *Energy Economics* 108: 105908. ISSN 0140-9883. <https://doi.org/10.1016/J.ENERCO.2022.105908>.
- Dong, K., X. Ren, and J. Zhao (2021). “How does low-carbon energy transition alleviate energy poverty in China? A non-parametric panel causality analysis.” *Energy Economics* 103: 105620. ISSN 0140-9883. <https://doi.org/10.1016/J.ENERCO.2021.105620>.
- Dong, K., F. Taghizadeh-Hesary, and J. Zhao (2022). “How inclusive financial development eradicates energy poverty in China? The role of technological innovation.” *Energy Economics* 109: 106007. ISSN 0140-9883. <https://doi.org/10.1016/J.ENERCO.2022.106007>.
- Eggleston, S., L. Buendia, K. Miwa, T. Ngara, and K. Tanabe (2006). *2006 IPCC guidelines for national greenhouse gas inventories*. Volume 5. Institute for Global Environmental Strategies Hayama, Japan.
- Foster, V., J.-P. Tre, and Q. Wodon (2000). “Energy prices, energy efficiency, and fuel poverty.” Latin America and Caribbean Regional Studies Programme. Washington, DC: World Bank.
- Goldemberg, J., T.B. Johansson, A.K.N. Reddy, and R.H. Williams (1988). *Energy for a sustainable world*. John Wiley & Sons New York ISBN 0470209836.
- Golosov, M., J. Hassler, P. Krusell, and A. Tsyvinski (2014). “Optimal taxes on fossil fuel in general equilibrium.” *Econometrica* 82(1): 41–88.
- Heindl, P. (2014). “Measuring fuel poverty: General considerations and application to German household data.” Technical report.
- Herrero, S.T. (2017). “Energy poverty indicators: A critical review of methods.” *Indoor and Built Environment* 26(7): 1018–1031.
- Hills, J. (2011). “Fuel poverty: the problem and its measurement.” IEA (2002). “Energy and Poverty. In: World Energy Outlook 2002.”
- IEA (2010). “Energy poverty: How to make modern energy access universal.” In: World Energy Outlook 2010. IEA (2020). “World Energy Balances.”

- Legros, G., I. Havet, N. Bruce, S. Bonjour, K. Rijal, M. Takada, and C. Dora (2009). "The energy access situation in developing countries: a review focusing on the least developed countries and Sub-Saharan Africa." World Health Organization.
- Lewbel, A. and K. Pendakur (2009). "Tricks with Hicks: The EASI Demand System." *The American Economic Review* 99(3): 827–863. <https://doi.org/10.1257/aer.99.3.827>.
- Lewis, P. (1982). *Fuel poverty can be stopped*. National Right to Fuel Campaign.
- Li, K., B. Lloyd, X.-J. Liang, and Y.-M. Wei (2014). "Energy poor or fuel poor: What are the differences?" *Energy Policy* 68: 476–481. <https://doi.org/10.1016/j.enpol.2013.11.012>.
- Mayer, I., E. Nimal, P. Nogue, and M. Sevenet (2014). "The two faces of energy poverty: a case study of households' energy burden in the residential and mobility sectors at the city level." *Transportation Research Procedia* 4: 228–240.
- McCaig, B. and N. Pavcnik (2015). "Informal employment in a growing and globalizing low-income country." *American Economic Review* 105(5): 545–550.
- McCaig, B. and N. Pavcnik (2018). "Export markets and labor allocation in a low-income country." *American Economic Review* 108(7): 1899–1941.
- Mirza, B. and A. Szirmai (2010). "Towards a New Measurement of Energy Poverty: A Cross-Community Analysis of Rural Pakistan." UNU-MERIT Working Paper Series 024, United Nations University. Maastricht Economic and social Research and Training Centre on Innovation and Technology.
- Moore, R. (2012). "Definitions of fuel poverty: Implications for policy." *Energy Policy* 49: 19–26. <https://doi.org/10.1016/j.enpol.2012.01.057>.
- Munyanyi, M.E. and S.A. Churchill (2022). "Foreign aid and energy poverty: Sub-national evidence from Senegal." *Energy Economics* 108: 105899. ISSN 0140-9883. <https://doi.org/10.1016/J.ENERCO.2022.105899>.
- Nguyen, T.T., T.-T. Nguyen, V.-N. Hoang, C. Wilson, and S. Managi (2019). "Energy transition, poverty and inequality in Vietnam." *Energy Policy* 132: 536–548. <https://doi.org/10.1016/j.enpol.2019.06.001>.
- Nussbaumer, P., M. Bazilian, and V. Modi (2012). "Measuring energy poverty: Focusing on what matters." *Renewable and Sustainable Energy Reviews* 16(1): 231–243. <https://doi.org/10.1016/j.rser.2011.07.150>.
- Pachauri, S., A. Mueller, A. Kemmler, and D. Spreng (2004). "On measuring energy poverty in Indian households." *World Development* 32(12): 2083–2104.
- Pendakur, K. (2009). "EASI Made Easier."
- Pereira, M.G., M.A.V. Freitas, and N.F. da Silva (2011). "The challenge of energy poverty: Brazilian case study." *Energy Policy* 39(1): 167–175.
- Pérez, J., E. Bernal, and P. Rodríguez-Sánchez (2021). "Towards Use of Cleaner Fuels in Urban and Rural Households in Colombia: Empirical Evidence from 2010 to 2016." *The Energy Journal* 42(5). ISSN 1944-9089. <https://doi.org/10.5547/01956574.42.5.JPER>.
- Phimister, E., E. Vera-Toscano, and D. Roberts (2015). "The Dynamics of Energy Poverty: Evidence from Spain." *Economics of Energy & Environmental Policy* 4(1): 153–166. ISSN 2160-5890. <https://doi.org/10.5547/2160-5890.4.1.EPHI>.
- Romero, J.C., P. Linares, and X. López (2018). "The policy implications of energy poverty indicators." *Energy Policy* 115: 98–108. <https://doi.org/10.1016/j.enpol.2017.12.054>.
- Sagar, A.D. (2005). "Alleviating energy poverty for the world's poor." *Energy Policy* 33(11): 1367–1372. <https://doi.org/10.1016/j.enpol.2004.01.001>.
- Schuessler, R. (2014). "Energy Poverty Indicators: Conceptual Issues-Part I: The Ten-Percent-Rule and Double Median/Mean Indicators." *ZEW-Centre for European Economic Research Discussion Paper* (14-037).
- Sesan, T. (2012). "Navigating the limitations of energy poverty: Lessons from the promotion of improved cooking technologies in Kenya." *Energy Policy* 47: 202–210. <https://doi.org/10.1016/j.enpol.2012.04.058>.
- Singh, K. and R. Inglesi-Lotz (2021). "The Role of Energy Poverty on Economic Growth in Sub-Saharan African Countries." *Economics of Energy & Environmental Policy* 10(1). ISSN 2160-5890. <https://doi.org/10.5547/2160-5890.9.2.KSIN>.
- Tovar Reaños, M.A. and N.M. Wölfling (2018). "Household energy prices and inequality: Evidence from German microdata based on the EASI demand system." *Energy Economics* 70: 84–97. <https://doi.org/10.1016/j.eneco.2017.12.002>.
- Welsch, H. and P. Biermann (2017). "Energy Affordability and Subjective Well-Being: Evidence for European Countries." *The Energy Journal* 38(3): 159–176. ISSN 1944-9089. <https://doi.org/10.5547/01956574.38.3.HWEL>.
- WorldBank (2019). "World development indicators." Technical report.
- Zhao, J., Q. Jiang, X. Dong, and K. Dong (2021). "Assessing energy poverty and its effect on CO2 emissions: The case of China." *Energy Economics* 97: 105191. ISSN 0140-9883. <https://doi.org/10.1016/J.ENERCO.2021.105191>.



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